NARCISO III JAVIER

Computer Science Student • Systems, Backend & Software Engineering

Baguio City, Philippines | +63 976 451 1638 | renzoj156@gmail.com | linkedin.com/in/narcisoiii-javier | github.com/narcisoJavier | narcisojavier.vercel.app

EDUCATION

Saint Louis University *Baguio City, Philippines*

Bachelor of Science in Computer Science *Expected Graduation: June 2027*

- **Credentials & Certifications:** Smart City Challenge – Certified Participant (2024) • AI Development Track – Specialization Certificate
- **Relevant Coursework:** Data Structures & Algorithms, Operating Systems, Database Management Systems, Computer Networks, Software Engineering

TECHNICAL SKILLS

Languages:	Python, Go, C++, C#, Dart, JavaScript (ES6+), TypeScript, Node.js, PHP, SQL
Frameworks & Libs:	Unity 3D, Flutter, React, Next.js, PyQt6, Leaflet.js, Vite, Tailwind CSS
Tools & Infrastructure:	Docker, Docker Compose, Git, GitHub, Linux/Bash, VS Code Remote Dev Containers, Supabase, MySQL
Core Areas:	Systems Programming, Network Protocols (SSH, SFTP), GIS Mapping, Microservices, Desktop Automation, RESTful APIs

SELECTED TECHNICAL PROJECTS

Tether – *Mobile SSH & Mesh Terminal Client* *github.com/narcisoJavier/Tether*

Technologies: *Flutter, Dart, SSH/SFTP Protocols, VT100 Terminal Emulation, Tailscale Mesh Engine, Android Keystore*

- Architected an Android-first systems client combining a native Dart SSH/SFTP engine, VT100/xterm terminal emulation, and direct port forwarding.
- Integrated embedded Tailscale mesh networking to establish authenticated zero-config P2P connections to remote servers without public port forwarding.
- Implemented hardware-backed cryptographic storage via Android Keystore to securely safeguard private keys, passwords, and session credentials.

geoCradle – *Watershed & Administrative Boundary GIS Platform* *github.com/narcisoJavier/geoCradle*

Technologies: *React, Vite, Leaflet.js, GeoJSON, PMTiles, Supabase, Tesseract OCR Pipeline*

- Developed an interactive geospatial web application visualizing 13 major river basins and administrative boundaries across the Cordillera region.
- Engineered client-side vector-tile rendering using PMTiles and GeoJSON for 60 FPS drill-down navigation and high-density topographical layer toggling.
- Integrated a Supabase backend with an automated OCR extraction pipeline to ingest and catalog administrative boundary survey metadata.

Campus Navigator CS312 – *Containerized Microservices Routing Engine* *github.com/narcisoJavier/Campus-Navigator*

Technologies: *Go, Node.js, PHP, Docker Compose, MySQL, Graph Algorithms, RESTful APIs*

- Architected a containerized microservices platform consisting of a Go routing engine, Node.js API gateway, PHP management service, and MySQL database.
- Implemented Dijkstra's shortest-path algorithm in Go with custom heuristics for campus accessibility rules, wheelchair bypasses, and role permissions.
- Designed a thread-safe in-memory graph cache with read/write mutex synchronization to handle concurrent pathfinding queries with sub-millisecond latency.

MultiTask ContextSwitch – *Desktop Workflow & Window Focus Automator* *github.com/narcisoJavier/MultiTask_ContextSwitch*

Technologies: *Python, PyQt6, Windows Win32 APIs, QSystemTray, Desktop Automation Hooks*

- Developed a Windows desktop workflow automation tool that monitors real-time generation states in web-based AI tools via OS-level window handles.
- Implemented global keyboard hooks via Win32 APIs, instant target-window focus switching, system-tray minimization, and session cache cleanup.

ADDITIONAL TECHNICAL PROJECTS

Hand Sign Recognition CNN – *Computer Vision Prototype* *Google Colab Notebook*

Technologies: *Python, OpenCV, CNN, NumPy, Google Colab*

- Built a convolutional neural network prototype performing real-time webcam frame preprocessing, feature extraction, and multi-class gesture classification.

OpenCode DevContainer Setup – *Reproducible Tooling & Environment* *github.com/narcisoJavier/OpenCode-VSCode-Setup*

Technologies: *Docker, VS Code Remote Dev Containers, Linux/Bash, Security Hardening*

- Designed a reproducible development environment featuring containerized toolchains, non-root security isolation, and automated configuration.